Urban Taxes: January Update

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Literature on the effect of property taxation on home values

- The theoretical literature on local public finance says that the effect of property taxes on home values should be strictly negative, as long as property taxes are capitalized (either fully or partially). This was first noted by Oates (1969).
- Under full capitalization, a property that has a finite life span of n years will have a market value P_t in year t $(1 \leq t \leq n)$ given by

$$P_t = \sum_{s=t}^{n} \frac{Y_s - \tau P_s}{(1+i)^{s-t}}$$

  where Y_s is the inflow of property value in year s. i is the interest rate and τ the property tax rate.
- We can see that P_t decreases with τ: a property tax should have a negative effect on prices.

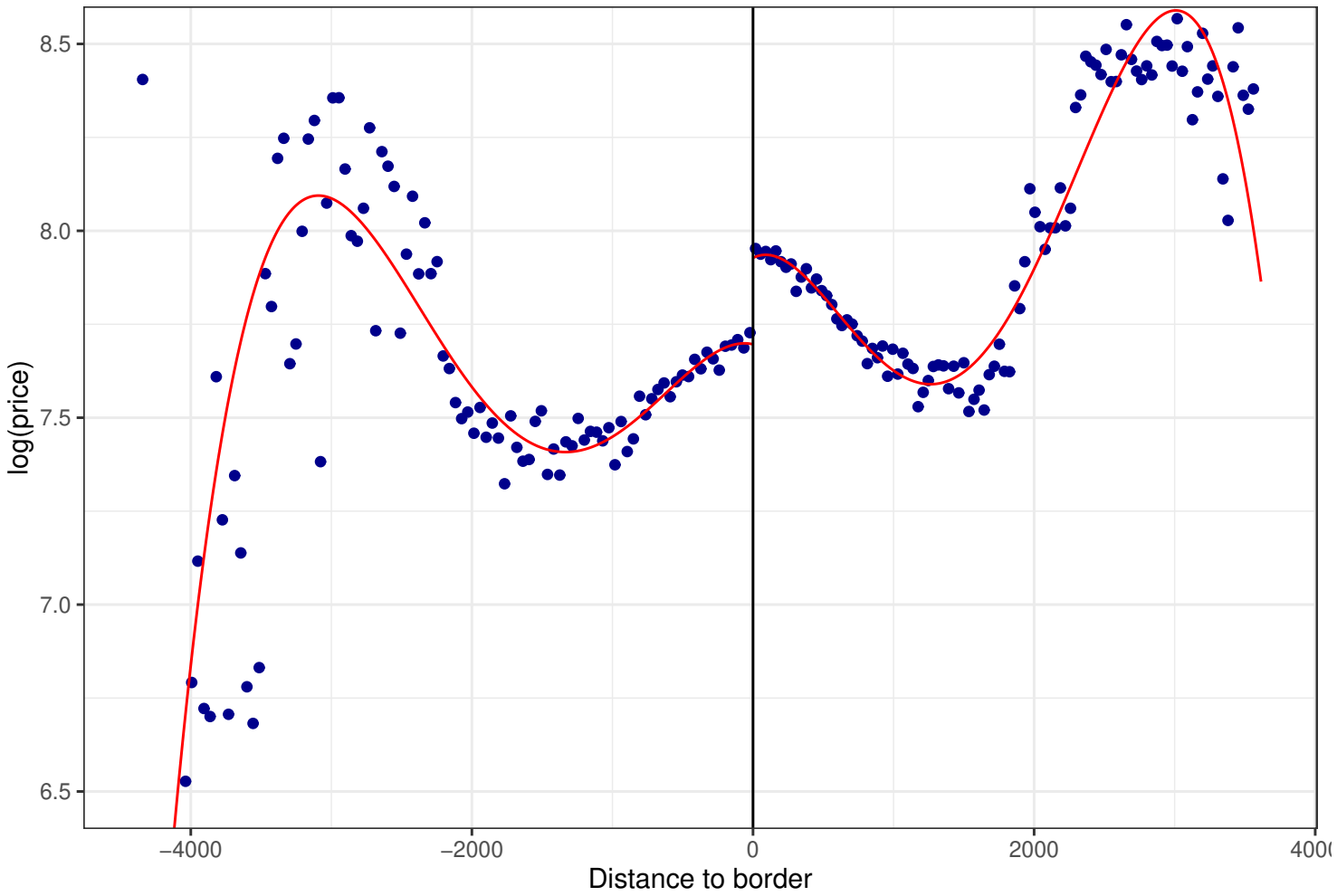
Literature on the effect of property taxation on home values

- Nonetheless, according to empirical studies, there is no consensus on the degree of tax capitalization:
  - Bai et al. (2014) studies two cities in China, finding that a property-tax experiment lowered the Shanghai average home price by 11%–15% but raised the Chongqing average home prices by 10%–12%.
  - They argue that the price increase in Chongqing can be driven by a spillover effect from high-end to low-end properties.
  - Elinder and Persson (2017) show house prices did not respond to a substantial cut in the national property tax in Sweden. However, effects are heterogeneous, as observe price increases only in a small segment of the market containing properties with very high tax values.
  - Oliviero and Scognamiglio (2019) show that a one standard deviation increase in municipal-level property tax intensity leads to a 2.7% reduction of property values in the year of an Italian tax reform.
  - See Hilber (2017) for a review of the literature up to 2017.
- Still, finding a positive link between property taxation and home values is rare and requires good justification.

Identification in our paper

- In our setting, properties above a certain tax appraisal threshold set by the government are subject to property taxes, while those below it do not pay.
- Moreover, in January 2013 the Chilean IRS introduced Homogeneous Areas (HA), which homogeneize the unit-value appraisal of a given square meter. In this sense, two identical homes located at different HAs with different unit-values will face a different per-square-meter-tax rate.
- Thus, the idea is to compare neighboring homes which are extremely close but located at different HAs. Intuitively, these homes *should* be extremely similar in many characteristics (observable and unobservable), and so the only difference in prices between them *should* be due to the tax differentials.
- This approach is called a geographic regression discontinuity design (geographic RD). For example, Berger et al. (2016) use this approach to estimate the effect of border differentials in TV license fees on evasion.

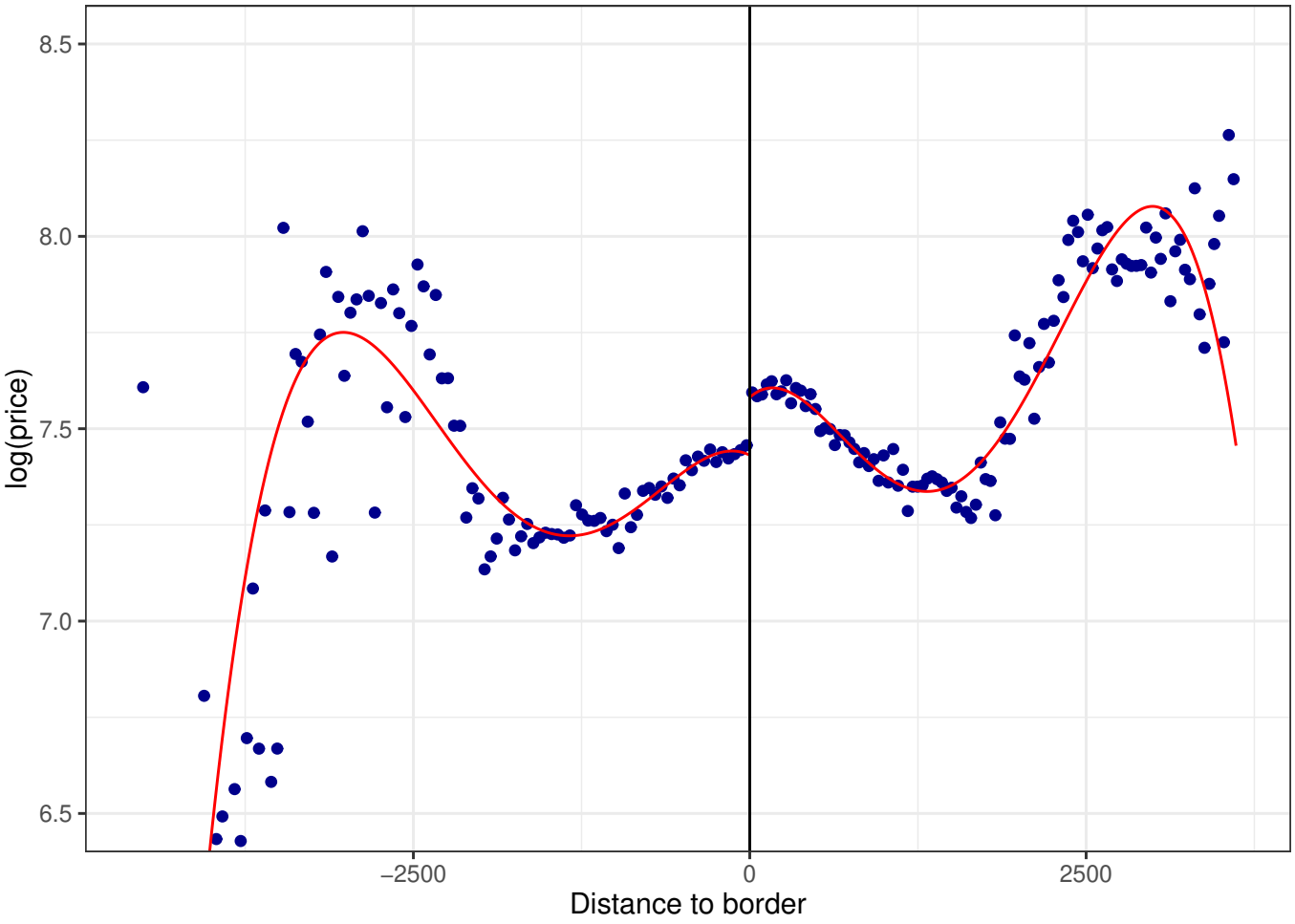
RD results: 2013-2017 home prices



RD results: 2013-2017 home prices

- Apparently, the effect of the moving from a low-tax-HA to a neighboring high-tax-HA is (strongly) positive, i.e., the effect of property taxation on home values is positive.
- Can we really trust these estimates?
- Since HAs were first established in 2013, running this same regression but using home prices observed before 2013 should not yield any significant results.

RD results: 2009-2012 home prices



RD results: 2009-2012 home prices

- The effect of the moving from a low-tax-HA to a neighboring high-tax-HA is also positive for the pre-treatment period.
- If the post-treatment RD is valid the pre-treatment regression should not show any significant results. Thus, we should not trust the previous conclusion.
- Why is this happening?:
  - Laws and institutions may change discontinuously (i.e. compound treatment) at the border cutoff.
  - People may chose to sort on either side of the borders (i.e. sorting around cutoff), leading to important differences between units in close geographic proximity even in the counterfactual world without treatment.
  - There might be administrative sorting, i.e., the government designs borders of HAs such that they reflect differences in home values, local public goods, amenities, etc...

Boletín SII

Noticias Sii

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Ante el CDE, municipalidades y Subsecretaría de Desarrollo Regional Santiago, 16 de Agosto de 2012

SII detalló metodología técnica utilizada para reavalúo 2013

El Servicio de Impuestos Internos (SII) efectuó el martes 14 de agosto, una presentación acerca del proceso de reavalúo de las propiedades no agrícolas que entrará en vigencia el próximo 1 de enero del año 2013, ante representantes del Consejo de Defensa del Estado (CDE), la Asociación de Municipalidades y la Subsecretaría de Desarrollo Regional.

La reunión detalló la metodología técnica que se está aplicando para definir los valores de los predios a aplicar en el próximo reavalúo, basado en un nuevo estudio urbano que permitió zonificar el territorio en Áreas Homogéneas (AH). Ello corresponde a sectores de las ciudades con características similares en cuanto a uso, categoría constructiva, nivel de densificación y normativa urbana, para los cuales es factible establecer un único valor de suelo, establecido en base al método comparativo de mercado.

Respecto de las construcciones, se expuso la clasificación que efectúa el SII para su incorporación en el catastro, y las modificaciones que se están realizando en los instrumentos que permiten definir su calidad. La valoración de las construcciones se hace sobre los costos de reposición, montos sobre los cuales se aplican los ajustes establecidos en el número 2° del Artículo 4° de la Ley N°17.235.

Asistieron a esta presentación el Presidente del Consejo de Defensa del Estado, Sergio Urrejola, el secretario ejecutivo de la Asociación Chilena de Municipalidades, Enrique Tapia Padilla, y asesores; y el jefe de la División Municipalidades de la Subsecretaría de Desarrollo Regional, Mauricio Cisternas, junto a asesores.

Por el SII, el director Julio Pereira, el subdirector de Avaluaciones, Ernesto Terán, y asesores.

Está contemplado que el SII se reúna próximamente con otros actores relevantes en el proceso, incluyendo a las municipalidades de mayor contribución al Fondo Común Municipal. Se destaca que el Servicio ya se ha reunido con muchas de éstas, explicando la metodología e intercambiando información relevante.

El reavalúo 2013 comprenderá la totalidad de las propiedades no agrícolas del país, alrededor de 5.700.000 bienes raíces.

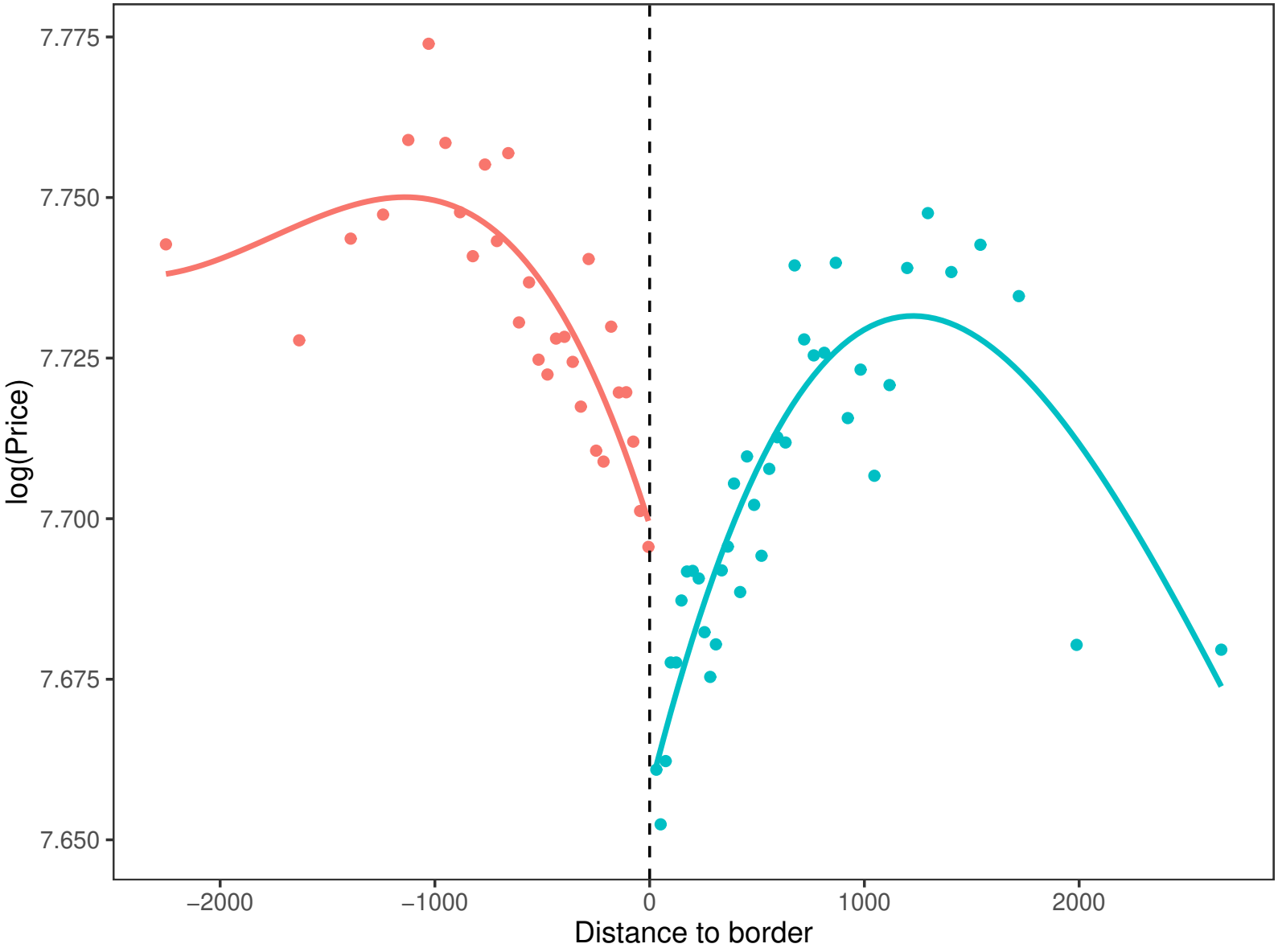
Challenges to identification

- The previous slide suggests that there is some administrative sorting going on: the government designed HAs in such a way that they reflect home value differentials.
- The IRS identifies high-value homes in an area, and then assignes them to high-unit-value HAs. This yields a significant pre-treatment effect. (This can also be interpreted as some form of reverse causality)
- Thus, when we move across the border cutoff, not only is the tax increasing, but also the quality of homes, and this is relfected in the price increase.
- Our objective is to uniquely identify the effect of the tax hike, and so we need to account for the differences in quality of the houses.

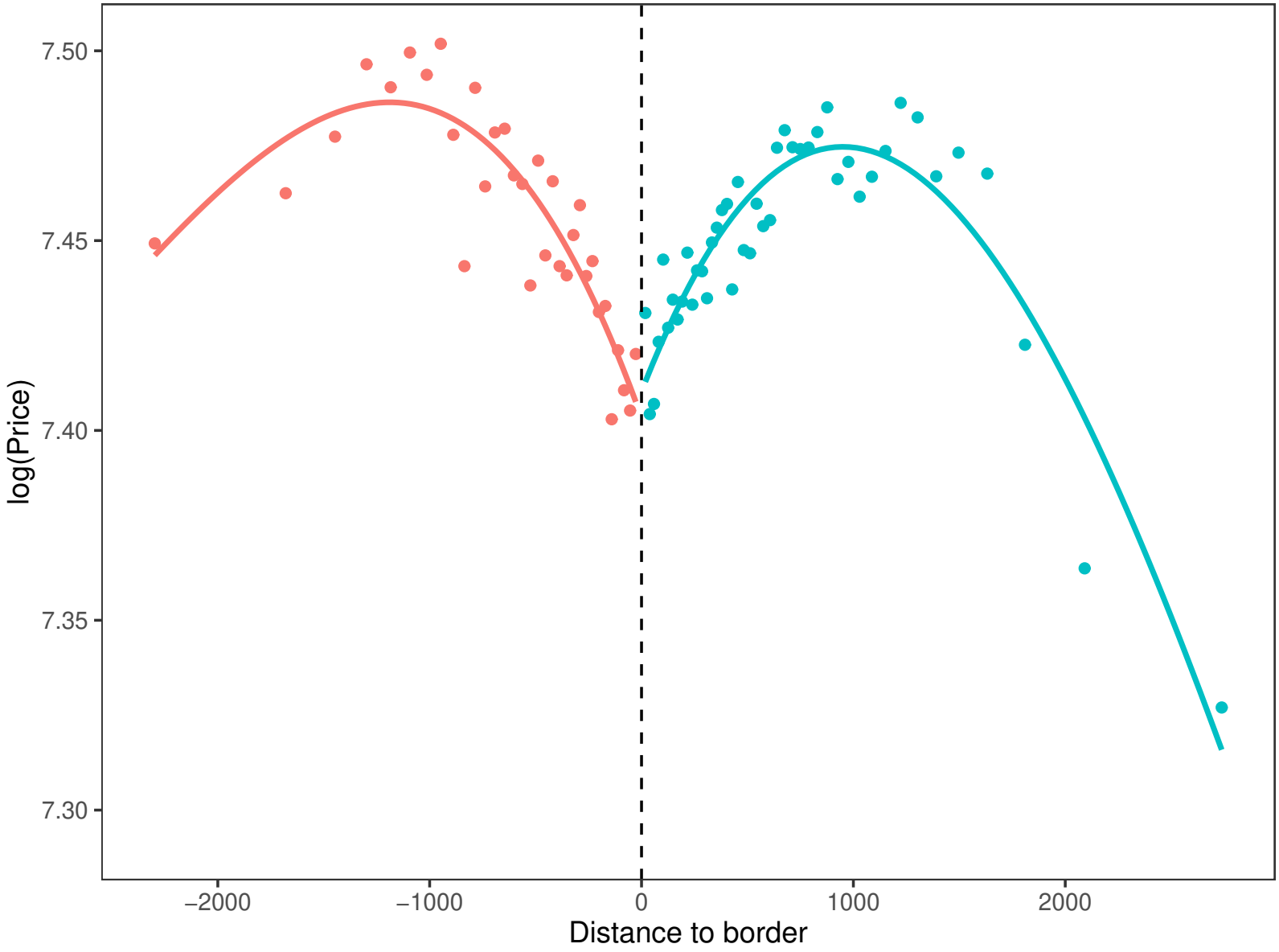
One possible solution

- One solution to some of these issues is to use a Differences-in-Discontinuities design (Diff-in-Disc), which removes all time-invariant sorting.
- Additionally, we need to control for the "quality" of the home. We can do this by using the quantile of the home's fiscal assessment.
- Hopefully, after doing this moving a low HA to a high HA will uniquely reflect the tax differential.

RD results with controls: 2013-2017 home prices



RD results with controls: 2009-2012 home prices



Discussion

- This looks much better. Let's see how the regression results compare.

	No controls		
	Post	Pre	Diff-in-Disc
Effect	0.1333775**	0.0544833	0.1032716***
Std. err	(0.05259518)	(0.05440086)	(0.01916054)
N	92825	69704	162529
	Fiscal assessment quantile as control		
	Post	Pre	Diff-in-Disc
Effect	-0.06184702**	-0.03698182	-0.035222546***
Std. err	(0.02620719)	(0.02950168)	(0.011121675)
N	92825	69704	162529

To-do list and challenges

- To-do list:
  - Use local polynomial regression to estimate the effects. From what I have estimated so far, the results should not change too much.
- Challenges:
  - We are having trouble estimating heterogeneus results (for example, by municipality), as this seems to break our current intuition.
- Once we solve this last issue, we should be ready to write.

References I

Bai, C.-E., Li, Q., and Ouyang, M. (2014). Property taxes and home prices: A tale of two cities. *Journal of Econometrics*, 180(1):1–15.

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